

How Localized Exposure to Inequality Mobilizes: Evidence from a Large Hospital Strike*

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Abstract

Mass labor strikes can make economic inequality publicly visible. Such visibility may be politically mobilizing when workers reveal how employment conditions shape the provision of an essential social good the community depends on. We explore this in the U.S. healthcare and social assistance sector, the dominant sector in recent strike activity, where worker demands frequently connect employment conditions to staffing and patient care. We document this pattern using data on U.S. labor actions since 2021 ($n = 4,755$), then examine the political effects of prolonged exposure to a multi-month nurses' strike in Massachusetts. Using voter-file and precinct-level election data and a difference-in-differences design comparing midterm elections (2022 vs. 2018), we assess effects on registration, turnout, and support for a redistributive ballot measure. We find no evidence that the strike increased turnout among existing voters near the hospital, nor support for the measure. But the share of registrations within one mile of the hospital was higher in 2022 than 2018, robust to demographic controls and absent in placebo comparisons. This increase accumulated gradually during and after the strike, growing more pronounced after a July 2021 service cutback. Localized strike exposure may draw new participants into the electorate by making the relationship between employment conditions and essential care salient, without activating existing voters or shifting vote choice.

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1 Introduction

Labor strikes are among the most visible forms of collective action in American politics (Lipsky, 1968; Schattschneider, 1960; Gourevitch, 2018; Amenta et al., 2010). They disrupt production but also often disrupt the routines of surrounding communities, curtailing access to services and generating sustained public attention (Gruber and Kleiner, 2012; Naidu and Reich, 2018; Krueger and Mas, 2004; Mas, 2008). These consequences may be especially politically salient when a dispute over employment conditions also explicitly implicates the quality and availability of an essential service.

Labor conflict has attracted renewed attention since the 2018 “Red for Ed” teacher walkouts, which were followed by high-profile actions at Amazon, Starbucks, and healthcare institutions nationwide. This resurgence raises broader questions about the conditions under which collective action mobilizes surrounding publics. Do such conflicts draw previously disengaged residents into the electorate, activate existing voters, or change political preferences?

Recent empirical work on the political effects of strikes offers mixed evidence. Previous studies’ findings suggest that the political consequences of strikes depend on the types of workers involved, the services they provide, and the character of the disruption they impose (Hertel-Fernandez et al., 2021; Lyon et al., 2024; Hertel-Fernandez, 2024; Smith and Son, 2014). We build on this literature by examining whether residential proximity to prolonged strike activity affects voter mobilization.

We explore this question in private-sector healthcare and social assistance, where workplace conditions bear directly on the care surrounding communities receive. By connecting disputes over staffing and working conditions to patient care, healthcare strikes make organizational decisions that patients may ordinarily experience as fixed salient, while identifying the employer as responsible for them. Prolonged exposure through picket lines, media coverage, and changes in hospital operations may turn a diffuse conflict over a private employment relationship into a locally salient public issue with political effects on surrounding residents (Schmidt, 1993; Kane and Newman, 2019; Anderson, 2017; Hertel-Fernandez, 2020; Hacker et al., 2021; Mazumder and Yan, 2024; McCombs and Shaw, 1972; Gamson and Wolfsfeld, 1993; Gillion, 2012; Gillion, 2013; Huber and Arceneaux, 2007).

First, we document the broader landscape of recent labor conflict in the U.S. using data from the Labor Action Tracker (Kallas et al., 2026). We then test the effects of prolonged exposure using the 2021–2022 Saint Vincent Hospital nurses’ strike in Worcester, Massachusetts, the longest nurses’ strike in state history.

Using individual-level voter file data for Worcester, we test whether proximity to the strike affected three outcomes: (1) voter turnout, (2) voter registration, and (3) support for a 2022 ballot measure on redistribution. These outcomes distinguish mobilization into the electorate from the activation of existing voters and from movement in an ideologically progressive direction. We employ a difference-in-differences design comparing changes in behavior between residents living near the hospital and those living farther away, across the 2018 and 2022 midterm elections. For turnout, we compare this change with the corresponding change between the 2014 and 2018 midterm elections; additional placebo comparisons use a non-struck local hospital.

We find no evidence that the strike increased turnout among existing voters. Across multiple specifications and distance cutoffs, the difference-in-differences estimates are null or slightly negative, and placebo tests show similar patterns, suggesting that any observed differences reflect secular trends rather than strike-induced mobilization.

However, we find that the strike was associated with an increase in new voter registrations near the hospital. The share of new registrations from within one mile of Saint Vincent Hospital increased from 14.7% in 2018 to 19.4% in 2022, a difference of approximately 4.8 percentage points. This change is robust to demographic controls, month fixed effects, and is absent in placebo tests.

An examination of registration timing reveals gradual increase during the latter portion of the strike which remained elevated afterward, with the increase concentrated following the hospital’s service cutbacks in July 2021. This suggests that direct service disruption may have driven the change.

Finally, we find no evidence that proximity to the strike increased support for the 2022 ballot measure to increase taxes on high-earners in order to fund local public goods. The raw correlation between distance and vote share disappears once pre-treatment partisanship is controlled for, indicating that the spatial pattern reflects the pre-existing political composition of neighborhoods near the hospital rather than a strike-induced shift.

The findings suggest that strikes can expand the electorate by drawing new participants into the political system, even when they fail to mobilize existing voters or shift vote choice. The mechanism appears to operate through direct service disruption: residents who experienced the tangible consequences of the strike were more likely to register to vote, potentially acquiring a stake in political outcomes they had not previously held. These results contribute to a growing literature on the political effects of labor conflict, while highlighting an important but previously unexamined pathway.

The Saint Vincent strike is an informative case for several reasons. At 301 days, it was the longest nurses’ strike in Massachusetts state history, leading to sustained exposure. It was politically salient, as multiple national and local elected officials publicly backed the nurses. The strike ended in a clear union victory, as nurses won improved staffing ratios, pay increases, and guaranteed reinstatement, providing a definitive endpoint and a resolution that may itself have carried political meaning. Furthermore, the strike occurred in a discrete geographic location, enabling us to exploit variation in residential proximity to the hospital, and its timing, beginning in March 2021 and ending in January 2022 positions it between the 2020 and 2022 elections, allowing us to compare pre- and post-strike electoral behavior. These features make this case well-suited to test whether the mechanisms identified in prior work operate similarly when essential services are disrupted.

Healthcare also represents an understudied sector in this literature. Unlike teachers, who serve primarily families with children, hospitals serve a broad cross-section of the community, and healthcare workers enjoy unusually high public trust (Brenan, 2026). To our knowledge, this study provides the first evidence of the political consequences of a private-sector healthcare strike.

The paper proceeds as follows. Section 2 develops the theoretical motivation, situating strikes within the broader literature on focusing events and disruptive collective action. Section 3 provides descriptive evidence on the contemporary distribution of U.S. labor actions. Section 4 describes the SVH strike and presents descriptive evidence of its local political salience. Section 5 briefly describes the data. Section 5 presents the empirical strategy and results for each outcome. Section 6 discusses mechanisms, limitations, and concludes.

2 Theoretical Motivation

2.1 Focusing Events and Political Behavior

Scholars have long debated whether salient events affect political behavior. While policy preferences and partisan attachments are typically stable (Green et al., 2002), a growing body of research documents conditions under which disruptive events can shift attitudes and mobilize political action. Two findings emerge consistently from this literature: proximity to the event matters, and effects depend on how directly individuals are affected by the disruption (Latané, 1981).

Prior studies emphasize the ways in which sudden, attention-grabbing occurrences elevate issues onto the public agenda, terming these *focusing events* (Birkland, 1998). Mass shootings, for instance,

increase support for gun control among proximate residents by making the threat of gun violence more personally salient (Newman and Hartman, 2019), and this attitudinal shift can translate into votes for gun control-related ballot measures (Markarian and Newman, 2024). Yet these effects are often localized and ephemeral, dissipating with distance and time (Roman and Thompson, 2024; Downs, 1972; Birkland and Lawrence, 2009; Zaller, 1992).

A second set of literature examines whether protest and collective action mobilize broader publics. Proximity and exposure remain central predictors. Enos et al. (2019) find that the 1992 Los Angeles Rodney King riot caused a marked liberal shift in policy support among nearby residents, with effects that persisted over a decade (Bobo et al., 1994). Reny and Newman (2021) demonstrate that the 2020 George Floyd protests decreased favorability toward police among politically liberal Americans, though attitudes among conservatives remained unchanged or polarized further. Wasow (2020) shows that the nature of the disruption itself shapes public response, as peaceful civil rights protests generated sympathy and political support, while violent unrest shifted attention toward law and order.

Across both sets of literature, salient, disruptive events appear to alter political behavior, but effects are conditional on geographic proximity, the nature of exposure, and individuals' prior attitudes.

Disruption may, however, be demobilizing. Material hardship can reduce the time, security, and social resources on which political participation depends (Wolfinger and Rosenstone, 1980; Rosenstone and Hansen, 1993; Brady et al., 1995; Verba et al., 1995; Schlozman et al., 2012; Leighley and Nagler, 2007). Hospital closures, for example, decrease turnout among nearby rural residents, with the largest reductions among those most vulnerable to the loss of care (Cox et al., 2025). Disruption may therefore act as a negative resource shock.

2.2 Strikes as Disruptive Public Events

Strikes share features with both focusing events and protest, as they are relatively sudden, visible, and symbolically charged (Birkland, 1998; Wasow, 2020). But they also differ in important ways. Unlike protests, which are typically directed at the state or the broader public (Lipsky, 1968; Schattschneider, 1960), strikes impose direct costs on third parties, such as consumers, patients, and students, who depend on the disrupted services. This raises the possibility that strikes mobilize not only through solidarity with workers but also through the experiential disruption of services on which community members rely (Hertel-Fernandez et al., 2021).

This third-party dimension of strikes has received less attention. When scholars have examined strikes specifically, they have traditionally understood them as instruments of economic bargaining: workers withdraw their labor to secure higher wages or improved conditions (Rubin, 1986; Burns, 2011). In this view, strikes are workplace-centered tactics designed to advance the interests of workers within a specific bargaining unit (Ahlquist and Levi, 2013).

Recent work has begun to challenge this narrow framing. Hertel-Fernandez et al. (2021) show that the 2018 public-sector teacher strikes increased support for unions and interest in labor action among exposed parents, suggesting that strikes can shape attitudes beyond workers themselves (Fantasia, 1988). Hertel-Fernandez (2024) extends this research to grocery worker strikes, finding increased public support for unions but limited spillover to related policy attitudes. Yet these studies focus on attitudinal outcomes. Whether strikes translate into downstream political behavior among exposed publics remains unclear. We argue that reconceptualizing strikes as public political events expands not only the set of actors we should expect them to influence but also the set of outcomes we should examine, including voter registration and electoral participation (Brady et al., 1995; Verba et al., 1995; Schlozman et al., 2012; Macdonald, 2019; Leighley and Nagler, 2007;

Feigenbaum et al., 2019; Han, 2014).

Hertel-Fernandez et al. (2021) find that the effect of teacher-strike exposure appeared to operate through perceptions of “linked fate,” or a sense that teachers’ struggles were connected to parents’ own interests, as well as more straightforward information provision (Dawson, 1994; Gay et al., 2016; Ahlquist and Levi, 2013). By contrast, the grocery-store strike studied in Hertel-Fernandez (2024) failed to engender similar perceptions of linked fate, which may explain its more limited effects. Lyon et al. (2024) provide the most comprehensive study of behavioral outcomes to date, examining the effects of teacher strikes on voter turnout using county-level data and individual-level survey responses. They find that teacher strikes decreased turnout by approximately three percentage points, with effects concentrated among Republicans. Parents, however, became more likely to turn out to vote. They interpret this as evidence that disruption can demobilize as well as mobilize.

These findings suggest that the political effects of strikes depend on features of the workers, the services they provide, and the nature of the disruption. They also raise questions about heterogeneity across contexts. Public-sector teacher strikes primarily affect families with school-age children and unfold in a context where partisan divisions over government spending may be salient (Lyon et al., 2024). Healthcare strikes, by contrast, affect a broader cross-section of the community and can pit workers against private employers rather than the state.

Moreover, existing work has focused almost exclusively on whether individuals who are already registered turn out to vote. Strikes may also draw new participants into the electorate. If strikes impose tangible costs on community members who previously had little stake in political outcomes, these individuals may register to vote for the first time. This pathway would be obscured in analyses focused on turnout alone.

The existing findings point to two mechanisms through which strikes might shape political behavior:

1. **Solidarity.** Observers identify with striking workers and are mobilized through a sense of shared fate. This mechanism should produce political effects when workers successfully frame their demands in publicly resonant terms or when political elites publicly endorse the strike.
2. **Service disruption.** Individuals who depend on the struck services experience tangible costs that give them a personal stake in the conflict. This mechanism should produce political effects when services are actually curtailed and should be concentrated among those who use the services rather than those who merely observe the picket line.

These mechanisms generate different expectations about the timing and geography of political effects. If solidarity drives mobilization, effects should emerge following high-profile political endorsements or media coverage. If disruption drives mobilization, effects should emerge when services are scaled back and should be concentrated in the service area of the struck institution rather than in its immediate physical vicinity.

2.3 Healthcare as a Distinctive Site of Mobilization

Existing research on the political effects of strikes focuses on teachers and retail workers. Healthcare remains understudied despite being central to the labor movement: nearly one in five private-sector union members worked in healthcare and social assistance in 2024 (U.S. Bureau of Labor Statistics, 2026). Furthermore, healthcare strikes may be particularly likely to generate political effects for two reasons. First, hospitals serve a broad cross-section of the community on an ongoing basis. Unlike schools, which affect primarily families with children, or grocery stores, which consumers can substitute, hospital disruptions are tangible and widespread. Second, nurses

consistently rank as the most honest and ethical profession (Brenan, 2026). A strike that pits nurses against a for-profit hospital corporation (Schumacher et al., 2026) may activate latent sympathies in ways that other labor actions are less likely to.

2.4 Theoretical Expectations

We test three hypotheses:

H1 (Turnout): *Residents near the struck hospital will turn out at different rates in the post-strike election than residents farther away, relative to pre-strike elections.*

Prior work offers competing expectations. Teacher strikes decrease turnout, particularly among Republicans (Lyon et al., 2024), suggesting disruption demobilizes. But healthcare workers enjoy higher public trust, and the COVID-19 pandemic may have heightened sympathy for nurses, potentially generating mobilization instead. We remain agnostic and treat turnout as an empirical question.

H2 (Registration): *The share of new voter registrations from near the struck hospital will increase in the post-strike period, relative to the pre-strike period.*

Existing work focuses largely on turnout among registered voters. Strikes may also operate on the extensive margin, drawing new participants into the electorate. Residents who experience service disruptions acquire a personal stake in political outcomes they may not have had before. Registration is therefore the outcome most closely connected to political incorporation (Meredith, 2009).

H3 (Vote choice): *Residents near the struck hospital will be more supportive of a redistributive ballot measure in the post-strike election, relative to residents farther away.*

If exposure generalizes from a workplace conflict to broader views of economic inequality, proximity may also increase support for redistribution. This is a more demanding expectation than registration because a strike against a private employer provides no automatic partisan cue and the ballot measure is not directly about healthcare or labor rights.

The three outcomes distinguish entry into politics, activation within the existing electorate, and the direction of political response. These hypotheses could yield divergent results. The timing of any changes helps adjudicate mechanisms, because we expect that changes that emerge after political endorsements point to solidarity. If changes emerge after service cutbacks, disruption is the more plausible causal pathway.

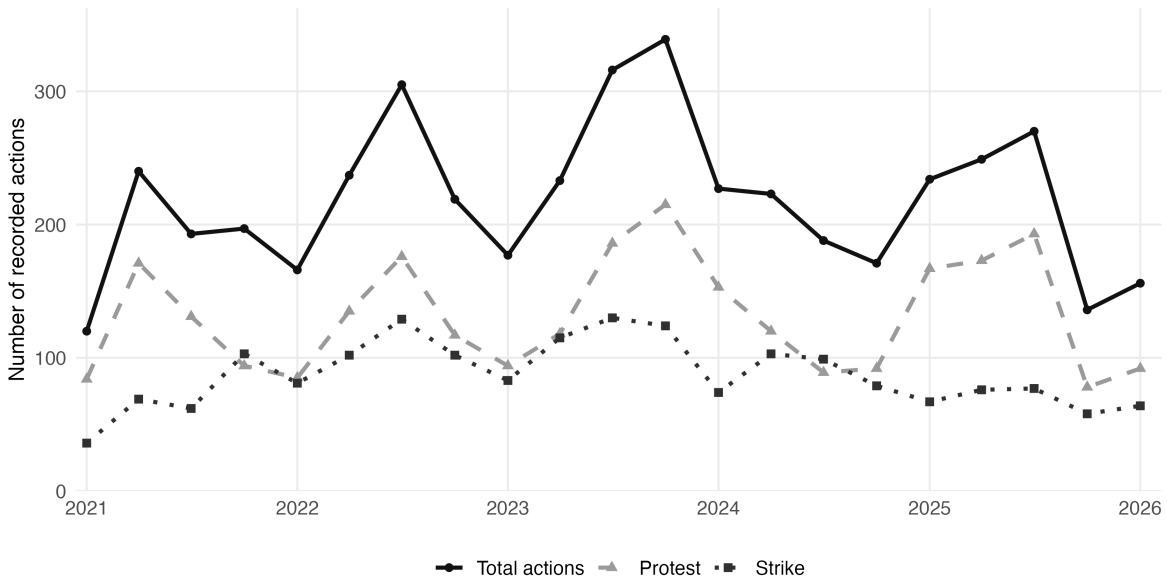
3 Labor Action in the Contemporary United States

We use data from the Labor Action Tracker (LAT) to provide a descriptive overview of recent strikes and labor protests across public- and private-sector employers (Kallas et al., 2026). Unlike the Bureau of Labor Statistics series restricted to major work stoppages, the tracker includes actions of all reported sizes. It begins in 2021, so we use it to characterize the recent composition of labor conflict rather than a long-run historical trend.

The June 2026 extract contains 4,755 labor actions, comprising 1,874 strikes and 2,881 protests. Figure 1 plots quarterly counts of strikes, protests, and total recorded actions. Protests outnumber strikes in most quarters, and both types of action exhibit substantial quarter-to-quarter variation.¹

¹The final point is the first quarter of 2026 and is not a completed annual count.

Labor strikes and protests in the United States since 2021
 Quarterly actions recorded by the Labor Action Tracker; through Q1 2026



Notes: Strikes and protests are shown separately rather than pooled.
 The LAT records actions below the 1,000-worker threshold used in federal major-work-stoppage statistics.

Figure 1: Recorded U.S. Labor Actions by Quarter, 2021–2026. Quarterly counts distinguish strikes and labor protests and show their combined total. Data for 2026 include only the first quarter.

Figure 2 reports counts of strikes and protests across the largest industry categories. Recorded labor action is concentrated in service-providing sectors: the three largest categories are healthcare, education, and accommodation and food services. Healthcare accounts for 874 actions, or 18.4% of the full dataset, making it the most prevalent sector in the LAT. Healthcare labor conflict is therefore not confined to a few large national disputes, and is a recurring feature of the recent labor-action landscape.

Labor actions by industry, 2021–2026

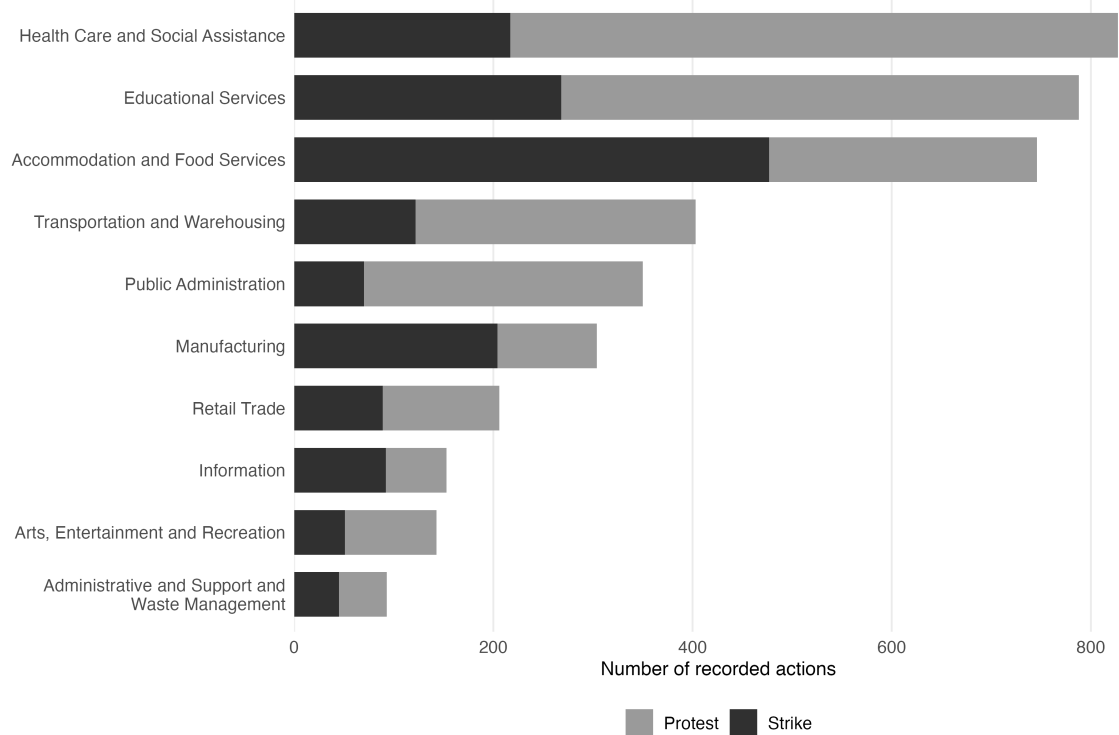


Figure 2: Labor Actions by Industry, 2021–2026. Counts distinguish work stoppages (strikes) from labor protests. The figure displays the largest recorded industry categories in the Labor Action Tracker. Categories include actions at public- and private-sector employers.

The demands made in these disputes show why healthcare strikes extend beyond the employment relationship. We use a dictionary-based, multi-label coding scheme to classify each action’s stated demands into non-mutually-exclusive substantive categories. Panel A of Figure 3 shows the prevalence of these categories separately for healthcare strikes and protests. Among healthcare strikes, 69.5% include wage demands, 64.6% include staffing or patient-care demands, and 24.3% concern working conditions. Panel B then groups healthcare strikes according to whether they contain employment demands, service-provision demands, both, or neither. A majority combine employment and service provision, while only 12.4% concern service provision without an identified employment demand.

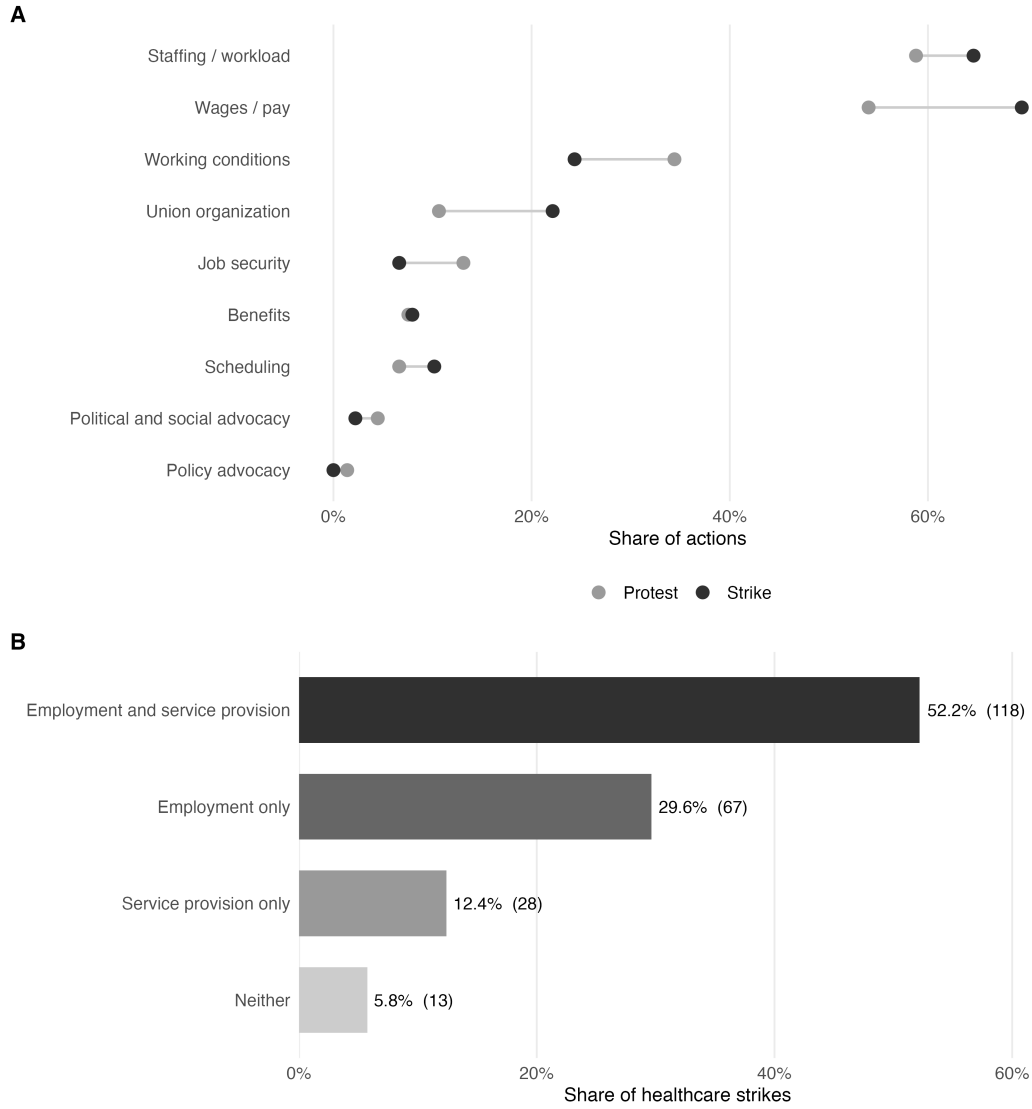


Figure 3: Demands in Healthcare Labor Actions. Panel A shows the share of healthcare strikes and protests containing each non-mutually-exclusive demand category ($N = 226$ strikes and $N = 648$ protests). Panel B classifies healthcare strikes into mutually exclusive combinations of employment and service-provision demands. Employment demands include compensation, benefits, job security, union organization, and scheduling. Service-provision demands include staffing, patient safety, and related working conditions.

The overlap is substantively important. When healthcare workers connect compensation, retention, or workplace authority to staffing and patient safety, they describe employment conditions as inputs into care. The strike thereby addresses two audiences at once: the employer that controls the workplace and the patients and residents who depend on the service. The Saint Vincent case is thus representative of broader healthcare-industry demands that link employment conditions to service provision. Its extended duration also makes the Saint Vincent strike especially useful for investigating the political effects of strike exposure: over 301 days, residents had recurring opportunities to encounter the conflict.

4 Background

4.1 The Saint Vincent Hospital Nurses’ Strike

Saint Vincent Hospital is a 297-bed acute care hospital located in Worcester, Massachusetts, owned by Tenet Healthcare, a for-profit hospital corporation headquartered in Dallas, Texas. The hospital’s approximately 800 registered nurses are represented by the Massachusetts Nurses Association (MNA) union.

On February 23, 2021, the MNA issued the required 10-day notice of its intention to begin an open-ended strike. In the days leading up to the walkout, support broadened. On February 28, environmental (i.e., janitorial) workers represented by the United Food and Commercial Workers (UFCW), whose contract had expired, announced they would join the picket line in solidarity. The strike officially began on March 8, 2021, with approximately 700 nurses walking off the job and replacement nurses brought in to staff the hospital (Elk, 2021). This began the longest nurses’ strike in Massachusetts history.

Public and political support for the strikers grew quickly. On March 12, U.S. Senator Elizabeth Warren and Representative Jim McGovern joined nurses on the picket line, amplifying national attention to the dispute. At the local level, City Councilor Candy Mero-Carlson introduced a resolution before the Worcester City Council shortly thereafter expressing formal support for the striking nurses and other essential staff (Worcester Business Journal, 2021).

In public statements surrounding the resolution, council members framed the strike not simply as a labor dispute but as a matter of community moral obligation. As Councilor Mero-Carlson put it, the nurses had “always cared for this community” and had done so “throughout the worst public health crisis in our history, sacrificing their own health and the health of their own families to keep us safe” (Kotsopoulos, 2021). By describing the strike as a necessary action taken to secure adequate staffing and resources “to protect us,” local officials cast the nurses’ walkout as an extension of their caregiving role rather than a withdrawal from it. In this framing, the conflict with hospital management was positioned less as a private contractual disagreement and more as a public struggle over inequality, frontline sacrifice during COVID-19, and the community’s obligation to stand with healthcare workers.

As the strike stretched into the summer, its operational effects became more visible. On July 15, Saint Vincent scaled back certain services, making the consequences of the conflict tangible to patients and families throughout the region. Negotiations continued through the fall. After more than nine months on strike, the parties reached a tentative agreement on December 17, 2021. Nurses approved the agreement on January 3, 2022, formally ending the strike.

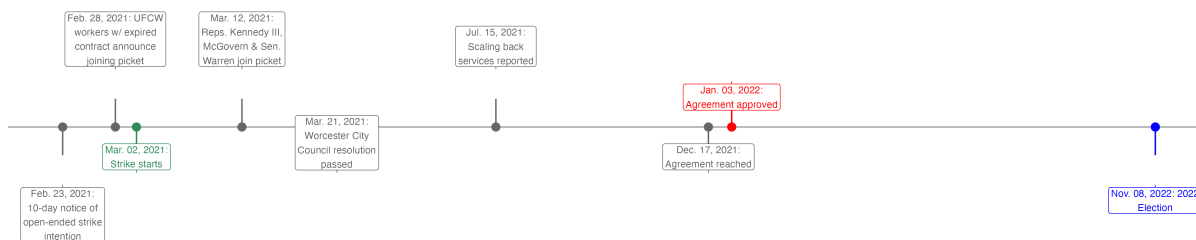


Figure 4: Timeline of Key Strike-Related Events. The strike began on March 8, 2021 and ended on January 3, 2022. Key events include political endorsements (March 12), the Worcester City Council resolution (March 21), the hospital’s service cutbacks (July 15), and the strike’s successful resolution (January 3, 2022).

4.2 Strike Salience and Persistence

For a strike to shape political behavior, we argue that it must command sustained public attention. Many forms of economic conflict remain background conditions of everyday life; salience requires disruption that concentrates attention and persists long enough to reshape political judgments (Birkland, 1998). Several features of the Saint Vincent strike suggest it met this threshold.

First, the strike unfolded in a space that community members regularly move through. Healthcare services are embedded in the routines of everyday life. Residents arriving for appointments, visiting family members, or commuting through the area would have encountered the picket line directly. When services were scaled back in July, normal patterns of care were disrupted.

Second, the timeline of events reveals a tight clustering of early strike-related activity that elevated the conflict from a sector-specific labor dispute to a matter of community-wide concern. Within two weeks of the strike’s onset, U.S. senators and representatives had joined the picket line, and the Worcester City Council had passed a formal resolution of support. A second wave of attention followed in July when the hospital announced service cutbacks. Rather than a single event that faded from view, the strike generated repeated moments of heightened visibility over its 301-day duration.

Third, descriptive measures of behavior and media attention confirm that residents were actively attending to the strike. Figure 5 displays Google search intensity for strike-related terms in Massachusetts. Searches for “Saint Vincent Hospital,” “Saint Vincent strike,” and “Worcester nurses strike” rose sharply following the strike’s onset, with subsequent spikes coinciding with the service cutbacks in July and the strike’s resolution in January 2022.

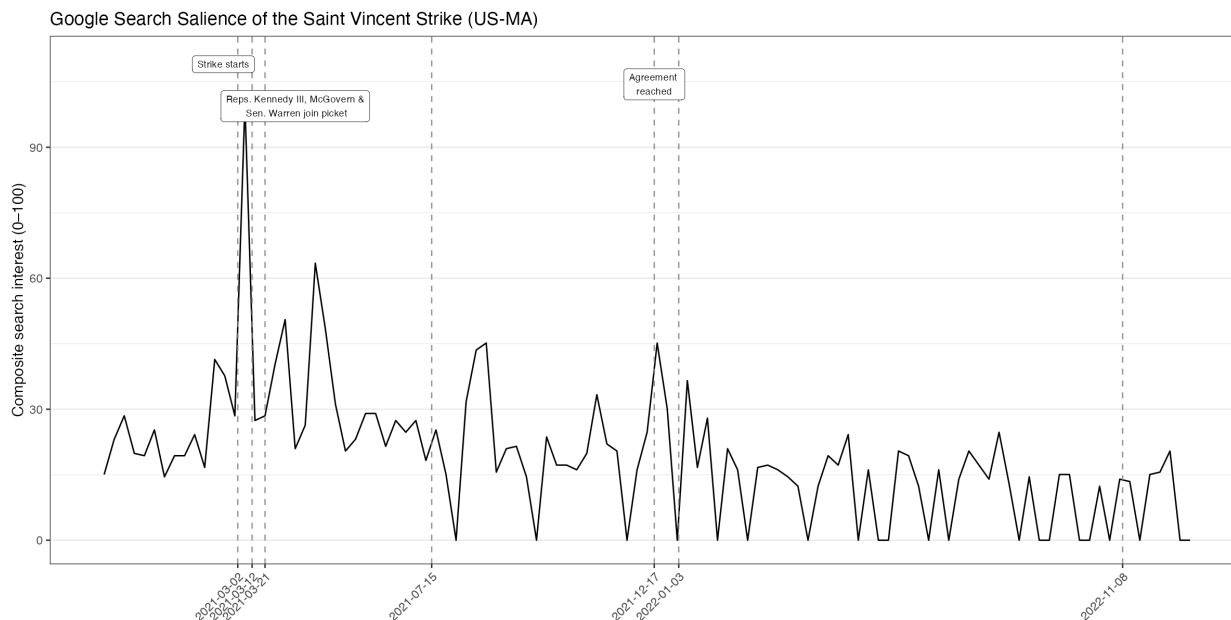


Figure 5: Google Search Interest in Saint Vincent Strike-Related Terms (Massachusetts). Searches for “Saint Vincent Hospital,” “Saint Vincent strike,” “St Vincent Hospital strike,” and “Worcester nurses strike” rise immediately following the onset of the strike on March 8, 2021. Major strike-related events are indicated by vertical dashed lines.

Local media coverage may have reinforced this attention. Figure 6 displays weekly counts of strike-related newspaper articles identified through ProQuest. Coverage increased sharply following the strike’s onset and spiked again around the July service cutbacks. Importantly, coverage persisted

between major events. The strike received steady attention throughout its duration, sustaining its presence in the local information environment preceding the 2022 election.

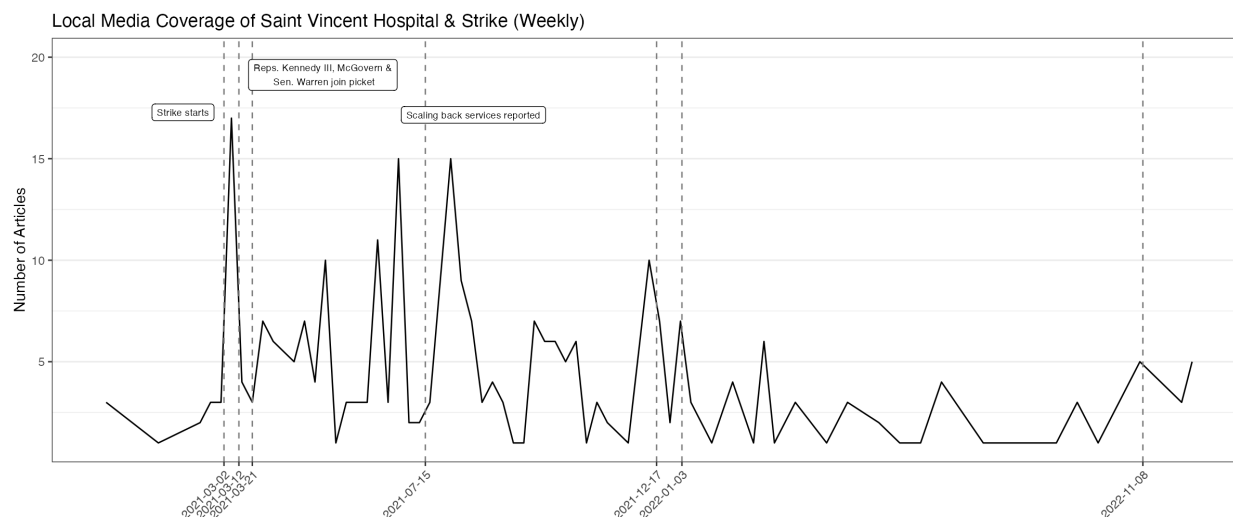


Figure 6: Local Newspaper Coverage of the Saint Vincent Hospital Strike. Weekly counts of strike-related newspaper articles, identified in ProQuest using the search terms “Saint Vincent Hospital,” “Saint Vincent strike,” and “Worcester nurses strike.” Coverage increases sharply following the strike’s onset on March 8, 2021, and spikes again around the service cutbacks (July 15, 2021). Vertical dashed lines denote major strike-related events.

Together, these patterns suggest that the Saint Vincent strike was not merely long but also salient and persistent. The concentration of early political endorsements, measurable spikes in information-seeking behavior, sustained media coverage, and tangible service disruptions suggest that the strike became a focal community issue during the period between the 2020 and 2022 elections. We next examine whether this heightened salience translated into measurable shifts in political behavior.

5 Data

5.1 Voter File

We use individual-level voter file data for Worcester, Massachusetts obtained from Statara Solutions, a commercial vendor. The voter file contains the universe of registered voters, including each voter’s name, residential address, registration date, party affiliation, validated turnout history for all elections from 2000 to 2025, and geocoded residential latitude and longitude coordinates. We use these coordinates to calculate each voter’s distance to Saint Vincent Hospital.

For the turnout analysis, we use a fixed cohort of voters who were registered by 2006 and had non-missing demographic information, yielding an analysis sample of 28,728 voters. Holding the cohort constant allows us to compare the same voters across the longer series of midterm and presidential elections, although it necessarily focuses the analysis on long-term registrants. For the registration analysis, we identify new registrants by their registration date and restrict to those who registered in the January–October period of each comparison year to ensure comparability across election cycles.

The voter file includes several demographic variables: party registration (Democrat, Republican,

Unenrolled, other), gender, race (modeled), age, education level (modeled), and homeownership status (modeled). We use these variables as controls in our regression specifications.

5.2 Precinct-Level Election Returns

For the vote choice analysis, we use precinct-level returns for the 2022 Massachusetts general election, obtained from the Secretary of the Commonwealth. We focus on Question 1, the “Fair Share Amendment,” which imposed an additional 4% tax on income over one million dollars. We also obtain precinct-level returns for the other three 2022 ballot measures (Question 2 on dental insurance, Question 3 on alcohol licenses, and Question 4 on driver’s licenses) for use in placebo tests.

To control for pre-treatment partisanship, we estimate 2020 Biden vote share at the 2022 precinct level using areal interpolation from Census tracts obtained from the MIT Election Data and Science Lab (Voting and Election Science Team, 2020). Precinct-level demographic controls (percent white, percent age 65+, percent renter, median income, population density) are drawn from the 2020 American Community Survey via the same interpolation procedure (U.S. Census Bureau, 2021).

Worcester County contains 258 precincts; our analysis sample includes 256 precincts with complete data on all covariates.

6 Results

6.1 Study 1: Turnout

6.1.1 Empirical Strategy

To estimate the effect of strike exposure on turnout, we employ a difference-in-differences design comparing changes in turnout between 2018 and 2022 for voters living near versus far from the hospital:

$$(1) \quad Turnout_{it} = \beta_1(Near_i \times Post_t) + \beta_2Near_i + \beta_3Post_t + X_i'\gamma + \varepsilon_{it}$$

where $Near_i$ indicates residence within one mile of SVH, $Post_t$ indicates the 2022 election, and X_i is a vector of standard demographic controls including party registration, age, age squared, gender, and race. The coefficient β_1 captures the differential change in turnout for near-hospital voters relative to far-hospital voters. We estimate this model using logistic regression. We then apply the same specification to the 2014 and 2018 midterm elections and formally test whether the 2018–2022 differential change differs from the prior midterm trend.

Our empirical strategy differs from prior work on strikes and political behavior in several respects. We examine a single prolonged strike and exploit within-county variation in residential distance to the struck institution. This design holds constant many community-level confounders, such as local political culture, media environment, and labor market conditions, that may differ across struck and non-struck counties.

6.1.2 Results

Table 1 presents the difference-in-differences estimates for voter turnout. Between 2018 and 2022, turnout declined 4.1 percentage points more among voters within one mile of SVH than among voters farther away. The adjusted interaction estimate is negative and statistically significant

($\beta = -0.171$, $p < 0.001$). However, turnout had already declined relatively more near the hospital during the prior midterm period.

Table 1: Difference-in-Differences Estimates: Voter Turnout

Specification	Raw DiD	Estimate	SE	p -value
2018–2022	−4.1pp	−0.171	0.044	< 0.001
2014–2018 placebo	−1.6pp	−0.110	0.042	0.009
Difference between periods	−2.5pp	−0.062	0.073	0.392

Note: Estimates from logistic regression models. The fixed sample includes voters registered by 2006 in Worcester, MA ($N = 28,728$). Models include controls for party, age, age squared, gender, and race. The final row formally compares the near-versus-far change across the two midterm periods.

6.1.3 Placebo Test

We validate this design using a placebo test that applies the same specification to the 2014 and 2018 midterm elections, which occurred before the strike. This comparison holds election type constant and tests whether the near-versus-far turnout gap was already changing before the strike.

Turnout declined relatively more among voters living within one mile of Saint Vincent Hospital between 2018 and 2022. However, turnout had already declined relatively more near the hospital during the prior midterm period. A formal comparison of the two changes is not statistically significant ($\beta = -0.062$, $p = 0.392$), indicating that the post-strike decline cannot be distinguished from the pre-existing geographic trend. We therefore find no detectable evidence that the strike affected turnout among existing voters.

6.2 Study 2: Registration

6.2.1 Empirical Strategy

For the registration analysis, we test whether the near-hospital share of new registered voters increased between 2018 and 2022. Specifically, we compare the share of new registrants living within one mile of SVH in 2022 (post-strike) versus 2018 (pre-strike). We estimate:

$$(2) \quad Near_i = \alpha + \delta Post_t + X_i' \gamma + \varepsilon_i$$

where $Near_i$ is an indicator for whether registrant i lives within one mile of SVH, $Post_t$ indicates registration in 2022 (versus 2018), and X_i is a vector of demographic controls. The coefficient δ captures the change in the probability that a new registrant lives near the hospital. A positive δ indicates that the near-hospital share of registrations increased after the strike.

To examine timing, we plot the weekly share of registrations from within one mile of SVH and compare this share before and after key strike-related events.

We use individual-level voter file data rather than county-level aggregates or survey responses. This allows us to observe the timing of voter registration and describe whether changes appear abruptly around particular events or accumulate over the course of the strike. Because the events and forms of exposure overlap, the timing analysis does not separately identify their effects.

6.2.2 Results

As a first pass, Table 2 presents the share of new voter registrations coming from within one mile of SVH in each midterm year. The share increased from 14.7% in 2018 to 19.4% in 2022, a difference of 4.8 percentage points.

Table 2: New Voter Registrations by Distance to SVH

Year	N Registrants	Share Within 1 Mile
2018 (pre-strike)	2,967	14.7%
2022 (post-strike)	5,284	19.4%
Difference		+4.8pp

Note: Sample includes voters who registered January–October of each year.

Table 3 presents regression estimates with controls. The baseline estimate of 4.76 percentage points is attenuated to 3.11 percentage points after controlling for party, age, gender, race, education, and homeownership, indicating that roughly one-third of the raw difference reflects compositional changes in who registered. But the effect remains statistically significant ($p < 0.001$) and is robust to the inclusion of month fixed effects.

The 3.1 percentage point adjusted increase in registration share corresponds to approximately 160 more registrations within one mile of the hospital than the model predicts under the 2018 spatial distribution. This represents roughly 1% of the near-hospital electorate—a modest but non-trivial change concentrated in a small geographic area. We interpret this quantity as the magnitude of the post-strike geographic shift, rather than as a count of registrations definitively caused by the strike.

The placebo comparison of 2016 to 2020 yields a smaller estimate of 1.64 percentage points that is marginally statistically significant ($p = 0.054$). This reduces concern that the full 2022 increase simply continues a shift of the same magnitude, but it also counsels against treating the across-cycle comparison as a sharp experiment.

Table 3: Registration Difference-in-Differences Estimates

Model	Estimate	SE	p -value
Base (no controls)	+4.76pp	0.0087	< 0.001
With demographic controls	+3.11pp	0.0087	< 0.001
With controls + month FEs	+3.07pp	0.0088	< 0.001
Placebo (2016 vs. 2020)	+1.64pp	0.0085	0.054

Note: Dependent variable is an indicator for registering within 1 mile of SVH. Sample includes new registrants January–October of each year. Demographic controls include party, age, gender, race, education, and homeownership.

6.2.3 Descriptive Timing and Mechanism

A key question is what aspect of the strike might account for the geographic change in registration. We distinguish between two potential mechanisms: (1) *political salience*, in which media coverage and elite endorsements raise awareness and prompt civic engagement; and (2) *direct disruption*, in which tangible disruption to services creates a personal stake that motivates political action. The analyses in this subsection are descriptive. The strike’s events were not randomized, exposure

accumulated over time, and registration is highly seasonal; the timing cannot identify the causal effect of any single event.

Table 4 examines the timing of the registration effect by dividing the strike period into sub-periods defined by key events. The share of registrations from within one mile of SVH was essentially unchanged in the early months of the strike (15.5% vs. 15.9% pre-strike). The share increased modestly after the July 2021 service cutback (17.4%) and increased substantially after the strike’s successful resolution in January 2022 (19.4%).

Table 4: Registration Share by Strike Period

Period	N	Share Within 1 Mile
Pre-strike (through Feb 2021)	8,510	15.9%
Strike, pre-cutback (Mar–Jun 2021)	1,738	15.5%
Strike, post-cutback (Jul 2021–Jan 2022)	2,555	17.4%
Post-strike (Jan 2022+)	5,272	19.4%

To describe whether registration share near SVH changed around specific strike-related events, Table 5 reports mean registration shares in the weeks immediately before and after five key dates: the strike’s commencement (March 2, 2021), U.S. Senators and Representatives joining the picket line (March 12), the Worcester City Council resolution (March 21), the hospital’s cutback of inpatient services (July 15), and the strike’s resolution (January 3, 2022).

The period following high-profile political endorsements showed little change in near-SVH registrations (+0.6 percentage points, not statistically significant). The City Council resolution was followed by a decline in registration share (−3.4 percentage points), a pattern consistent with substantial week-to-week noise rather than a stable discontinuity.

The service cutback in July 2021 was followed by a 4.0 percentage point increase in near-SVH registration share (from 14.9% to 18.9% in the six weeks before and after). The formal discontinuity test does not reach statistical significance, however, and the series does not display a clean break. We therefore view the pattern as consistent with a gradual increase during the latter part of the strike, not as evidence that the cutback independently caused registration.

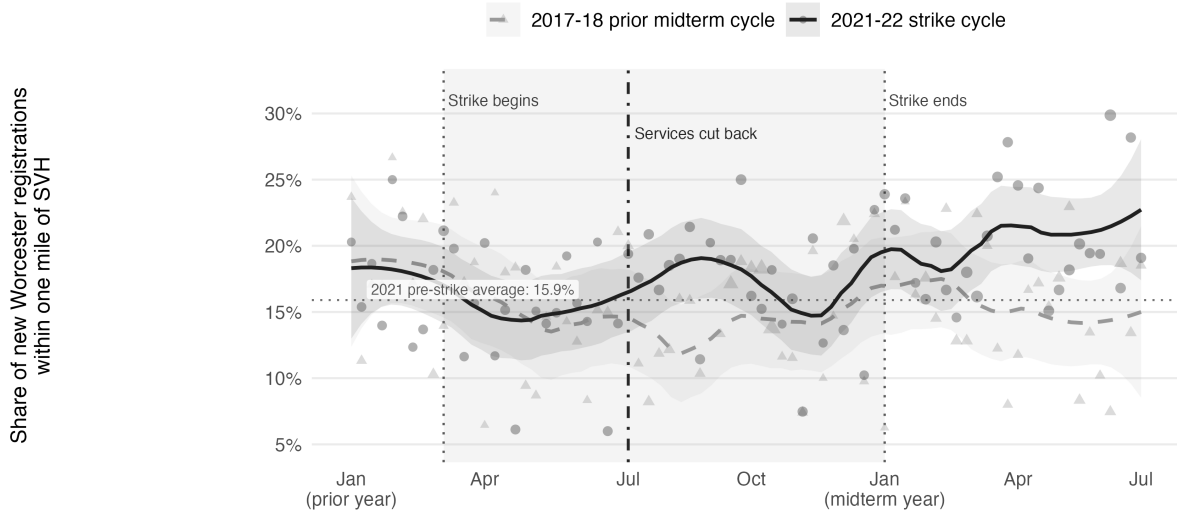
Table 5: Registration Response to Strike-Related Events

Event Type	Event	Pre Mean	Post Mean	Difference
<i>Political/Symbolic Events</i>				
	Politicians join picket (Mar 12)	16.3%	16.9%	+0.6pp
	City council resolution (Mar 21)	18.2%	14.9%	−3.4pp
<i>Disruption Events</i>				
	Strike begins (Mar 2)	16.6%	16.9%	+0.3pp
	Service cutback (Jul 15)	14.9%	18.9%	+4.0pp
	Strike ends (Jan 3)	18.2%	19.1%	+0.8pp

Notes: Pre and post means calculated using 4 weeks before/after for political events and 6 weeks before/after for disruption events. None of the formal discontinuity tests reach statistical significance at conventional levels due to week-to-week noise in the data.

A

Comparison with the prior midterm cycle



B

Estimates across distance thresholds and placebo comparisons

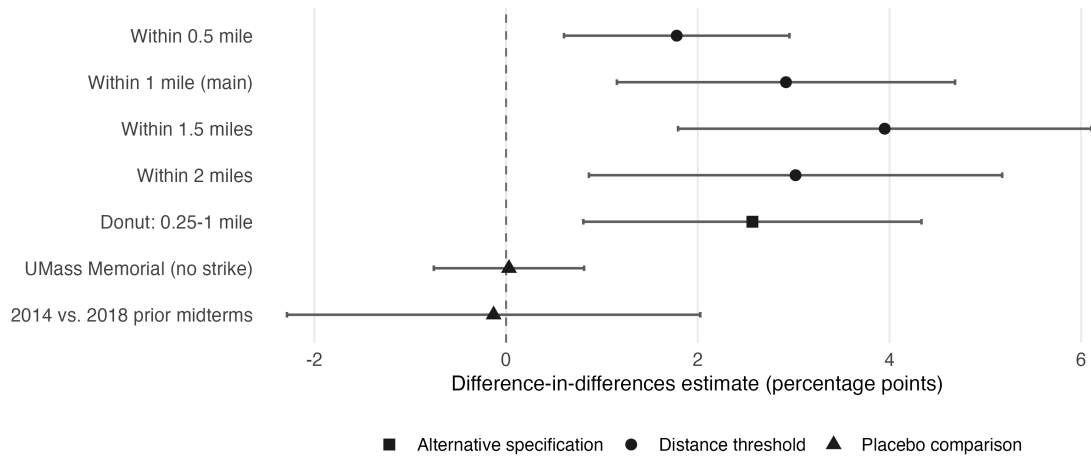


Figure 7: Registration Trends and Estimated Geographic Shift. Panel A compares the weekly share of new Worcester registrants living within one mile of Saint Vincent Hospital during the 2021–2022 strike cycle and the corresponding 2017–2018 midterm cycle. Point size reflects weekly registration volume; lines are LOESS fits with 95% confidence intervals. Shading marks the strike period, vertical lines mark the strike’s commencement, the July 2021 service cutback, and its resolution, and the horizontal dotted line shows the 2021 pre-strike average (15.9%). Panel B reports adjusted difference-in-differences estimates with 95% confidence intervals across distance thresholds, an alternative donut specification, and placebo comparisons. Models control for party identification, age, gender, race, education, and homeownership. Panel A is descriptive and does not identify the effect of any single event.

Table 7 (Panel A) shows that the increase in registration share is present across several distance thresholds but does not follow a strict decay pattern. The largest increase appears at 1.5 miles (+3.95 percentage points), with smaller increases both closer to the hospital (+1.78pp at 0.5 miles) and farther away (+3.02pp at 2 miles). This is not the monotonic gradient we would expect if exposure were determined only by direct sight of the picket line.

One possible explanation is residential composition. Table 6 shows that registrants within 0.5

miles had much lower homeownership rates in 2018 than those 1.5–2 miles away (16% versus 53%). If homeownership proxies residential stability or integration into local networks, residents somewhat farther from the hospital may have had more opportunities for repeated local discussion even while encountering the picket less directly. But this remains speculative. Renters may also experience substantial exposure, homeownership is an imperfect measure of community attachment, and we do not observe hospital use or residential mobility. The spatial pattern therefore shows that the registration shift extends beyond the hospital’s immediate blocks, not which form of exposure generated it.

Appendix Figure C1 plots the matched-cycle weekly trends separately for cumulative thresholds of 0.5, 1, 1.5, and 2 miles. The 2021–2022 series generally exceeds the prior-cycle series during the latter part of the observation period at each threshold, but the divergence is weakest and noisiest within 0.5 miles. The evidence therefore does not support a simple dose-response relationship in which political effects decline mechanically with straight-line distance from the picket line. We interpret proximity as locating registrants within the broader area surrounding the conflict, not as a direct measure of individual exposure intensity.

Table 6: Composition of New Registrants by Distance to SVH (2018)

	0–0.5mi	0.5–1mi	1–1.5mi	1.5–2mi	>2mi
N	135	300	511	442	1,579
Mean age	40.0	41.9	41.0	41.4	42.2
% Female	44.4	48.7	46.2	43.4	47.9
% Democrat	34.8	33.7	28.2	29.4	32.1
% Homeowner	15.6	22.3	32.7	53.4	57.7
% White	42.2	43.7	55.8	67.0	67.4

This interpretation is also consistent with results from a placebo comparison. Registrations did not increase near UMass Memorial Medical Center, a large hospital 2.3 miles from SVH that experienced no strike. The comparison reduces concern that the finding reflects a citywide shift common to neighborhoods surrounding major hospitals, although one untreated hospital cannot rule out every location-specific trend.

Taken together, the timing and spatial evidence is consistent with a geographically localized response to a prolonged, visible conflict over healthcare provision, but not with a simple distance-based exposure gradient. Registration did not jump following high-profile endorsements; its near-hospital share rose gradually during the latter portion of the strike and remained elevated afterward. The increase is larger beyond the hospital’s immediate blocks and is absent around the comparison hospital. Service disruption may have reinforced the strike’s salience, but the evidence does not cleanly separate disruption from repeated observation, local discussion, media coverage, strike duration, or resolution. We therefore treat these patterns as mechanism-consistent evidence rather than a causal adjudication between solidarity and disruption.

6.2.4 Robustness Checks

We conduct a series of robustness checks to validate the main registration finding. Results are summarized in Table 7.

Alternative Distance Cutoffs. The choice of one mile as the treatment boundary is somewhat arbitrary. To assess sensitivity, we re-estimate the main specification using alternative cutoffs of 0.5,

1.5, and 2 miles. The effect is positive and statistically significant at all cutoffs, ranging from 1.78 percentage points at 0.5 miles ($p = 0.001$) to 3.95 percentage points at 1.5 miles ($p < 0.001$). The consistency across specifications suggests the finding is not an artifact of the particular distance threshold chosen.

Placebo Hospital. As explained above, a key threat to identification is that the registration increase near SVH could reflect broader trends in central Worcester rather than the strike specifically. To assess this possibility, we conduct a placebo comparison using UMass Memorial Medical Center, a large hospital located about 2.3 miles from SVH that experienced no strike during this period. The estimated change in registration share within one mile of UMass Memorial is effectively zero: 0.03 percentage points ($p = 0.936$). This finding is inconsistent with a shift common to the areas around both major hospitals, though it cannot eliminate trends unique to the Saint Vincent neighborhood.

Prior Midterm Cycle Placebo. We also examine registration patterns in the prior midterm cycle (2014 vs. 2018). The estimated effect is -0.13 percentage points ($p = 0.908$), statistically indistinguishable from zero. This provides no evidence of an analogous shift in the immediately preceding midterm comparison, although a single prior cycle does not establish that neighborhood trends would otherwise have remained parallel in 2022.

Donut Specification. One concern is that the registration effect could be driven by hospital employees who registered near their workplace. To address this, we estimate a “donut” specification that excludes registrants living within 0.25 miles of the hospital. The effect remains positive and significant: 2.57 percentage points ($p = 0.003$).

Continuous Distance. Finally, we test whether the pattern appears using continuous distance rather than binary treatment indicators. We regress each registrant’s distance to SVH on the post-strike indicator and controls. Consistent with the threshold specifications, 2022 registrants lived an estimated 0.08 miles closer to SVH than 2018 registrants ($p = 0.001$).

Table 7: Robustness Checks: Registration Effect

Specification	Estimate	SE	p-value
<i>Panel A: Distance Cutoffs</i>			
0.5 mile	+1.78pp	0.006	0.001
1 mile (main)	+2.92pp	0.009	0.001
1.5 miles	+3.95pp	0.011	<0.001
2 miles	+3.02pp	0.011	0.008
<i>Panel B: Placebo Tests</i>			
UMass Memorial (no strike)	+0.03pp	0.004	0.936
2014 vs. 2018 (prior midterms)	−0.13pp	0.011	0.908
<i>Panel C: Alternative Specifications</i>			
Donut (0.25–1 mile)	+2.57pp	0.009	0.003
Continuous distance	−0.081 mi	0.025	0.001
Log distance	−0.056	0.014	<0.001

Notes: All models include controls for party, age, gender, race, education, and homeownership. Sample includes new registrants January–October of each comparison year. Panel A shows the estimated increase in the share of registrations from within the specified distance of SVH in 2022 vs. 2018. Panel B shows placebo tests: UMass Memorial is a nearby hospital with no strike; 2014 vs. 2018 tests for pre-existing trends. Panel C shows alternative specifications: the donut excludes registrants within 0.25 miles; continuous distance shows the change in mean distance to SVH.

Taken together, these robustness checks strongly support the interpretation that the Saint Vincent Hospital nurses’ strike plausibly caused an increase in voter registrations among nearby residents. But did the strike merely bring new people into the electoral process, or did it have a more durable influence on how residents perceived inequality, potentially shaping their political responses when given the opportunity to act on those perceptions? To address this question, we turn to examining whether proximity to the strike was associated with vote choice on a ballot measure related to redistribution.

6.3 Study 3: Vote Choice on Redistributive Ballot Measure

6.3.1 Empirical Strategy

Next, to test whether proximity to the strike affected vote choice, we examine precinct-level support for Question 1, the “Fair Share Amendment,” a 2022 ballot measure imposing an additional 4% tax on incomes over one million dollars. We estimate:

$$(3) \quad Q1Support_p = \alpha + \beta Distance_p + X_p' \gamma + \varepsilon_p$$

where $Q1Support_p$ is the share of yes votes in precinct p , $Distance_p$ is the distance from the precinct centroid to Saint Vincent Hospital, and X_p is a vector of precinct-level controls including demographic composition (percent white, percent age 65+, percent renter, median income, population density) and partisanship. We estimate specifications with post-treatment partisanship (2022 Healey gubernatorial vote share) and pre-treatment partisanship (2020 Biden vote share, estimated via areal interpolation from Census tracts) to assess whether any distance gradient reflects strike-induced attitude change or pre-existing partisan geography.

6.3.2 Results

Table 8 presents the results. In the base specification with no controls, distance from Saint Vincent Hospital is negatively associated with Question 1 support: each additional mile is associated with 0.42 percentage points lower support ($p < 0.001$). This raw correlation is consistent with the hypothesis that strike exposure increased support for redistribution.

However, the relationship attenuates substantially with the inclusion of demographic controls. Adding precinct-level demographics reduces the coefficient by approximately 80%, to 0.09 percentage points per mile ($p = 0.03$). The association is null once we control for partisanship. With the inclusion of 2022 Democratic vote share, the distance coefficient flips sign and fails to reach statistical significance ($\beta = 0.04$, $p = 0.16$). This pattern holds when we substitute pre-treatment partisanship (2020 Biden vote share) for post-treatment partisanship: the distance coefficient remains near zero and non-significant ($\beta = 0.04$, $p = 0.32$).

Table 8: Distance from Saint Vincent Hospital and Question 1 Support

	(1) Base	(2) Demographics	(3) + 2022 PID	(4) + 2020 PID
Distance (miles)	-0.0042*** (0.0006)	-0.0009* (0.0004)	0.0004 (0.0003)	0.0004 (0.0004)
Demographic controls		✓	✓	✓
2022 Healey share			✓	
2020 Biden share				✓
N (precincts)	258	258	238	256
R^2	0.22	0.68	0.89	0.79

Notes: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Robust standard errors in parentheses. Demographic controls: % white, % age 65+, % renter, median income, population density.

The similarity of results with pre- and post-treatment partisanship rules out the concern that controlling for 2022 vote share absorbs the relationship. If the strike had shifted both Question 1 support and broader partisan orientation, the distance coefficient should remain significant with 2020 Biden but not 2022 Healey. Instead, the coefficient is null in both specifications, indicating that the raw correlation reflects the pre-existing political composition of neighborhoods near the hospital.

6.3.3 Placebo: Other Ballot Measures

As a further check, we examine whether distance predicts support for other 2022 ballot measures. Table 9 presents the distance coefficient from models controlling for demographics and pre-treatment partisanship across all four ballot questions. Distance does not significantly predict support for any measure: not Question 1 (millionaire tax), Question 2 (dental insurance), Question 3 (alcohol licenses), or Question 4 (driver's licenses). The absence of a distance gradient across measures unrelated to healthcare or redistribution reinforces the interpretation that the raw correlation reflects partisan geography rather than strike-specific effects.

Table 9: Distance from Saint Vincent Hospital and Support for 2022 Ballot Measures

	Q1 Millionaire Tax	Q2 Dental Insurance	Q3 Alcohol Licenses	Q4 Driver’s Licenses
Distance (miles)	0.0004 (0.0004)	0.0003 (0.0004)	−0.0004 (0.0003)	−0.0007 (0.0004)
<i>p</i> -value	0.35	0.47	0.19	0.12

Notes: All models include demographic controls and 2020 Biden vote share. N = 256 precincts. Robust standard errors.

In sum, we find no evidence that proximity to the strike increased support for redistributive taxation. The spatial pattern of Question 1 support is fully explained by the pre-existing partisan composition of neighborhoods near the hospital. While the strike may have expanded the electorate through new registrations, it did not shift vote choice among those who cast ballots.

7 Discussion

The Saint Vincent Hospital nurses’ strike lasted 301 days, drew support from national political figures, and ended in a decisive union victory. Yet its electoral consequences were uneven. New voter registrations increased near the hospital, while turnout among existing registrants was unchanged and support for a redistributive ballot measure did not differ from the rest of the city. The sections below examine what might explain this pattern.

7.1 Why Registration but Not Turnout?

The divergence between registration and turnout likely reflects differences in the margins of political participation. Registration captures entry into the electorate, while turnout captures participation among those already registered. These behaviors may respond to different forms of political stimulus.

The observed pattern is consistent with the strike drawing previously disengaged individuals into the electorate without mobilizing those already registered. Repeated visibility through picketing, worker claims, coverage, operational changes, and eventual resolution may have made a local distributive conflict politically relevant to some nonregistrants. Such diffuse exposure may be less suited to activating turnout, which often depends on direct mobilization through canvassing or targeted outreach.

The descriptive timing is consistent with accumulated exposure rather than an immediate reaction to the March 2021 political endorsements. Registration rose during the latter portion of the strike, after services had also been curtailed, but the available data do not isolate service disruption from other features of a conflict that had by then been visible for months.

The null turnout result also aligns with Lyon et al. (2024), who find that teacher strikes *decreased* turnout, particularly among Republicans. We find no evidence of either mobilization or demobilization by this metric. This may reflect differences between healthcare and education contexts: teacher strikes directly affect families with school-age children, a group with strong prior political identities, whereas hospital strikes affect a broader and more heterogeneous population. Alternatively, the null result may simply reflect insufficient statistical power to detect modest effects in a single-city study.

7.2 Why No Effect on Vote Choice?

The null finding on Question 1, the “millionaire tax” ballot measure, is perhaps more surprising. If the strike activated concerns about economic inequality or strengthened support for redistributive policies, we might expect residents near the hospital to vote more favorably on a progressive tax measure.

Three explanations are most plausible. First, the strike may have increased political participation without shifting policy preferences. New registrants may have had diverse ideological positions. Second, the link between the strike and state tax policy may have been too abstract. We expected the strike to make economic inequality salient and thereby increase support for redistribution. In reality, the strike’s focus was specifically on staffing ratios, wages, and workplace conditions at a single hospital. Voters may not have connected the two.

Third, the raw correlation between proximity to the hospital and Question 1 support was fully explained by pre-existing partisan geography. Precincts closer to Saint Vincent Hospital were more Democratic-leaning, as measured by 2020 Biden vote share. Once we account for this baseline partisan composition, proximity to the strike adds no additional explanatory power. This suggests that any relationship between strike exposure and vote choice operates through channels such as residential sorting or neighborhood-level political culture that predate the strike itself.

This null result (Hassell et al., 2020; Hassell and Holbein, 2024) contrasts with Markarian and Newman (2024), who find that proximity to mass shootings increases support for gun control ballot measures. The difference may lie in the directness of the policy connection. Gun control ballot measures offer voters an opportunity to respond directly to the threat made salient by a shooting. Question 1, by contrast, addressed a policy domain only loosely connected to the strike’s core issues. Future research might examine whether strikes have stronger effects on vote choice when ballot measures directly address labor rights, healthcare policy, or collective bargaining.

7.3 Accumulated Exposure and Service Disruption

The timing evidence shows that registration did not increase sharply after a single political endorsement or operational event. Its near-hospital share rose gradually during the latter portion of the strike and remained elevated afterward. This pattern fits an accumulation account in which repeated encounters make a conflict progressively more salient rather than producing an immediate response.

But accumulation is not a separate observable treatment. It encompasses visual exposure to the picket, conversations with workers and neighbors, local coverage, political endorsements, operational changes, and knowledge of the strike’s persistence and resolution. The July service cutback may have intensified this process by making the conflict more tangible. But because it occurred after months of other exposure and does not coincide with a statistically distinguishable break, we cannot attribute the registration pattern specifically to service disruption. The evidence supports treating disruption as a possible component of accumulated exposure, not as the identified mechanism.

The spatial pattern likewise narrows but does not resolve the possibilities. The registration increase was not largest in the immediate vicinity of the hospital (within 0.5 miles), where residents are disproportionately renters, but in the 1–2 mile ring. This is consistent with effects extending into the surrounding community rather than operating only through direct sight of the picket line. Homeownership may proxy for residential stability and embeddedness in local networks, but it does not establish greater exposure, and we lack data on hospital utilization and residential mobility. The result should therefore be read as a non-monotonic neighborhood pattern in need of explanation, not as evidence for a service-area mechanism.

7.4 Limitations

This study has several limitations. First, we examine a single strike in one city, limiting external validity. The Saint Vincent strike was long, visible, and successful, while shorter or less prominent strikes may have different effects. The strike also ended in a decisive union victory, which may have contributed to the mobilization pattern we observe. Strikes that end in defeat or stalemate could produce different, or even opposite, effects.

Second, we lack attitudinal data and cannot directly test mechanisms. The gradual timing is consistent with accumulated exposure, but accumulation comprises several overlapping processes and cannot itself be observed in the voter file. Future research should pair behavioral data with surveys measuring direct observation, conversations with workers, service use, perceived responsibility, linked fate, and policy attitudes.

Third, the voter file demographic data is partially modeled (race, education) rather than self-reported, introducing potential measurement error. While these variables are widely used in applied research, they may attenuate estimated effects or introduce bias if modeling errors correlate with proximity to the hospital.

Fourth, we cannot identify healthcare workers or hospital patients in the data, preventing analysis of whether effects are concentrated among those with direct connections to the hospital. Such information would help distinguish workplace networks, patient experience, and neighborhood exposure.

Fifth, our vote choice analysis is limited to aggregate precinct-level data, which provides less statistical power than individual-level analysis and cannot account for within-precinct heterogeneity. Individual-level vote choice data, if available, would permit sharper tests.

7.5 Implications

The first implication concerns how political participation is measured. Research on the political consequences of strikes and hardship commonly begins with registered voters and asks whether they turn out. Our results suggest that this can miss an earlier behavioral response. A visible local conflict may make politics relevant enough for some residents to enter the electorate without supplying the targeted contact needed to mobilize existing voters. Registration, turnout, and vote choice should therefore be treated as distinct stages of political incorporation rather than interchangeable indicators of mobilization.

Second, the findings clarify the political importance of employment conditions in sectors that provide essential social goods. The LAT data show that healthcare strikes routinely connect compensation and workplace authority to staffing and patient care. This means that the politics of social provision does not occur only through legislation, public budgets, or program eligibility. It also occurs inside the organizations that deliver services, where employment decisions shape capacity and quality. Strikes can bring those organizational choices before a broader public by making visible the labor on which care depends.

Third, the comparison with hospital closures points toward a broader distinction between hardship alone and hardship accompanied by collective interpretation. A closure removes a resource and may reduce participation (Cox et al., 2025). A strike can also impose costs, but workers remain publicly present, identify a responsible decision-maker, and offer an account linking their interests to those of service users. The Saint Vincent evidence cannot establish that this organizational frame causes mobilization, but it motivates future comparison across analogous disruptions, such as hospital closures and healthcare strikes, school closures and teacher strikes, or transit cuts and transit-worker actions. Such comparisons could test whether organized claim-making changes the

political consequences of service loss.

Finally, translating strike-induced mobilization into partisan or policy support may require workers and unions to connect their demands explicitly to electoral politics. We find no shift in support for a redistributive ballot measure, and a conflict with a private hospital provides no simple partisan target. Residents may register because a local dispute makes politics newly consequential while drawing different conclusions about workers, management, or government. For unions, this distinction implies that public visibility alone may broaden participation without converting it into support for labor or redistribution. That additional step may require explicit voter outreach and a clearer connection between workplace demands and electoral politics.

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A Regression Tables

A.1 Data Summary

The statewide precinct dataset contains 2,256 precincts with complete vote and demographic data (zero missing observations after merging). The Worcester subsample contains 60 precincts; Springfield contains 64. Demographics are interpolated from 2020 ACS 5-year block-group estimates using areal weighting. All regressions use HC2 heteroskedasticity-robust standard errors (`estimatr::lm_robust`).

A.2 Table A1: Statewide Question 1 Regression

Table A1: Statewide Precinct-Level OLS: Question 1 Yes Share (HC2 Robust SEs)

	Estimate	Std. Error	<i>t</i>	<i>p</i> -value
(Intercept)	0.998	0.063	15.82	< 0.001
Distance (miles)	0.000	0.000	1.13	0.258
% White	-0.107	0.013	-8.28	< 0.001
% Age 65+	-0.087	0.041	-2.10	0.036
Log population	-0.050	0.008	-6.23	< 0.001
Pop. density (km ²)	5.15×10^{-6}	9.63×10^{-7}	5.35	< 0.001
% Renter	0.264	0.031	8.49	< 0.001
Median HH income	-3.84×10^{-7}	1.02×10^{-7}	-3.77	< 0.001
Median house value	2.60×10^{-8}	1.62×10^{-8}	1.60	0.110
Unemployment rate	-0.213	0.115	-1.86	0.064
<i>N</i>	2,256 precincts			
<i>R</i> ²	0.415			
Adj. <i>R</i> ²	0.413			
<i>F</i> -statistic	236.7 on 9 and 2,246 df, $p < 0.001$			

A.3 Table A2: Worcester Precinct Regression (No Partisan Control)

Table A2: Worcester Precincts OLS: Question 1 Yes Share, No Partisan Control (HC2 Robust SEs)

	Estimate	Std. Error	<i>t</i>	<i>p</i> -value
(Intercept)	0.350	0.274	1.28	0.207
Distance (miles)	-0.021	0.009	-2.35	0.023
% White	-0.042	0.055	-0.78	0.442
% Age 65+	0.038	0.162	0.23	0.817
Log population	0.031	0.032	0.95	0.345
Pop. density (km ²)	9.71×10^{-6}	6.09×10^{-6}	1.59	0.117
% Renter	0.070	0.143	0.49	0.630
Median HH income	6.53×10^{-8}	7.28×10^{-7}	0.09	0.929
Median house value	4.97×10^{-8}	1.34×10^{-7}	0.37	0.712
Unemployment rate	-0.252	0.223	-1.13	0.263
<i>N</i>	60 precincts (DF = 50)			
<i>R</i> ²	0.476			
Adj. <i>R</i> ²	0.382			
<i>F</i> -statistic	5.135 on 9 and 50 df, $p < 0.001$			

A.4 Table A3: Springfield Precinct Regression (Placebo, No Partisan Control)

Table A3: Springfield Precincts OLS: Question 1 Yes Share (Placebo), No Partisan Control (HC2 Robust SEs)

	Estimate	Std. Error	<i>t</i>	<i>p</i> -value
(Intercept)	0.608	0.196	3.09	0.003
Distance (miles)	−0.005	0.004	−1.08	0.285
% White	−0.013	0.054	−0.25	0.805
% Age 65+	−0.107	0.130	−0.82	0.413
Log population	−0.001	0.025	−0.05	0.963
Pop. density (km ²)	8.68×10^{-6}	3.78×10^{-6}	2.30	0.026
% Renter	−0.011	0.092	−0.12	0.908
Median HH income	-1.69×10^{-7}	3.92×10^{-7}	−0.43	0.668
Median house value	-8.90×10^{-8}	1.39×10^{-7}	−0.64	0.524
Unemployment rate	0.047	0.130	0.36	0.721
<i>N</i>	64 precincts (DF = 54)			
<i>R</i> ²	0.369			
Adj. <i>R</i> ²	0.263			
<i>F</i> -statistic	4.715 on 9 and 54 df, <i>p</i> < 0.001			

A.5 Table A4: Worcester and Springfield with Partisan Control (2022 Governor Share)

Table A4: Worcester and Springfield OLS with Partisan Control: Healey 2022 Share (HC2 Robust SEs)

	Worcester (<i>N</i> = 60)				Springfield (<i>N</i> = 64)			
	Parsimonious		Full		Parsimonious		Full	
	Est.	<i>p</i>	Est.	<i>p</i>	Est.	<i>p</i>	Est.	<i>p</i>
(Intercept)	−0.096	0.546	−0.331	0.130	0.340	0.058	0.282	0.116
Distance (miles)	−0.010	0.141	−0.009	0.223	−0.001	0.653	0.001	0.813
% White	−0.029	0.552	0.021	0.613	0.015	0.574	0.040	0.297
% Age 65+	−0.046	0.784	−0.186	0.147	−0.202	0.070	−0.139	0.204
Log population	0.025	0.205	0.038	0.093	0.005	0.799	0.004	0.840
Dem. share (2022)	0.761	< 0.001	0.836	< 0.001	0.312	< 0.001	0.319	< 0.001
<i>R</i> ²	0.755		0.795		0.453		0.507	
Adj. <i>R</i> ²	0.732		0.754		0.406		0.415	

Note: “Parsimonious” includes only % White, % Age 65+, log population, and Democrat share. “Full” adds population density, % renter, median household income, median house value, and unemployment rate.

A.6 Table A5: Worcester Descriptive Statistics by Distance Bin

Table A5: Worcester Precincts: Q1 Support by Distance from Saint Vincent Hospital

Distance bin	N	Mean Q1	SD Q1	Dem. share	% White	% Renter	Med. income
<0.75 mi	7	0.648	0.046	0.747	0.396	—	—
0.75–1.5 mi	17	0.624	0.046	0.721	0.434	—	—
1.5–2.5 mi	18	0.557	0.057	0.658	0.626	—	—
>2.5 mi	18	0.541	0.026	0.649	0.605	—	—

A.7 Table A6: Correlations with Distance from Saint Vincent Hospital (Worcester)

Table A6: Pairwise Correlations with Distance from Saint Vincent Hospital (Worcester, $N = 60$)

Variable	Correlation with distance
Q1 yes share	−0.606
Democrat share (2022)	−0.579
% White	+0.430
% Age 65+	+0.192
% Renter	−0.562
Median HH income	+0.570
Median house value	+0.479
Unemployment rate	−0.118
Pop. density	−0.636

Residual correlations with distance (after residualizing Q1 on 2022 Democrat share):

- Worcester: $r = -0.218$
- Springfield: $r = -0.047$

A.8 Table A7: Worcester County — Distance Coefficients Across Specifications

Table A7: Worcester County: Effect of Distance on Q1 Yes Share Across Specifications (HC2 Robust SEs)

Specification	Distance coefficient	p -value
No controls	−0.00418	< 0.001
With demographics	−0.00091	0.031
With demographics + 2022 PID	+0.00037	0.152
With demographics + 2020 PID	+0.00037	0.320

Note: Worcester County includes all 60 municipalities (257 precincts). Distance is measured from Saint Vincent Hospital (Worcester). 2020 PID is a pre-treatment Biden vote share interpolated via a 2020-precinct $\rightarrow$ census-tract $\rightarrow$ 2022-precinct areal crosswalk. Correlation between Biden 2020 and Healey 2022 county-level shares: $r = 0.847$.

A.9 Table A8: Worcester County by Distance Bin

Table A8: Worcester County: Q1 Support by Distance from Saint Vincent Hospital ($N = 257$)

Distance bin	N	Mean Q1	Dem. share	% White	% Renter
<2 mi	34	0.610	0.709	0.482	0.281
2–5 mi	40	0.519	0.619	0.684	0.141
5–10 mi	38	0.460	0.558	0.817	0.079
10–20 mi	96	0.446	0.518	0.828	0.110
>20 mi	49	0.471	0.521	0.808	0.128

A.10 Table A9: Q1 Support in Post-Industrial Massachusetts Cities

Table A9: Q1 Support in Post-Industrial Cities (all cities with ≥ 5 precincts)

City	Precincts	Q1 yes share	% White	% Renter	Med. income
Pittsfield	14	0.674	0.815	0.174	\$61,712
Lawrence	24	0.591	0.136	0.236	\$48,750
Holyoke	14	0.580	0.411	0.236	\$46,507
Lowell	32	0.573	0.497	0.208	\$60,408
Worcester	60	0.566	0.538	0.235	\$52,126
Springfield	64	0.563	0.304	0.197	\$42,079
Lynn	28	0.555	0.369	0.191	\$64,651
Brockton	28	0.533	0.292	0.145	\$65,601
Fitchburg	12	0.502	0.596	0.183	\$58,411
New Bedford	36	0.498	0.592	0.251	\$51,970
Chicopee	18	0.489	0.709	0.170	\$50,324
Fall River	27	0.482	0.738	0.284	\$48,091

A.11 Table A10: Worcester and Springfield — Actual vs. Demographically Predicted Q1 Support

Table A10: Actual vs. Predicted Q1 Vote Share: Worcester and Springfield

City	Actual Q1 share	Predicted (demographics)	Residual
Worcester	0.566	0.594	−0.028
Springfield	0.563	0.558	+0.004

Note: Predicted values from a precinct-level OLS of Q1 yes share on % white, % renter, % age 65+, median household income, and population density estimated on all 2,256 Massachusetts precincts.

B Turnout Analysis

B.1 Data Summary: Worcester County Voter File

The voter file contains 594,284 Worcester County registered voters with valid geocoordinates (deceased voters excluded). After restricting to voters registered by October 2020 with non-missing

age and party, the analysis sample is 394,150. Distance is measured in miles from Saint Vincent Hospital (42.2613°N, 71.8050°W) using the Haversine formula.

Table B11: Distance from Saint Vincent Hospital: Summary Statistics (Worcester County Voters)

	Min	Q1	Median	Mean	Q3	Max	<i>N</i>
Distance (miles)	0.006	5.22	12.27	12.41	17.96	356.15	594,284

Table B12: 2020 vs. 2022 General Election Turnout Crosstab (Worcester County)

	Did not vote 2022	Voted 2022
Did not vote 2020	179,698	15,168
Voted 2020	139,042	260,376

Table B13: Party Registration — Worcester County Voter File

Party	<i>N</i>
Democrat	111,461
Republican	59,056
Unenrolled	421,248
Libertarian	2,111
Green	408

B.2 Table A11: Turnout by Distance Band from Saint Vincent Hospital

Table B14: Voter Turnout by Distance Band from Saint Vincent Hospital (Worcester County)

Distance band	<i>N</i>	2020 turnout	2022 turnout	% Dem	% Rep	Mean age
0–0.5 mi	3,038	0.609	0.312	0.366	0.064	50.1
0.5–1 mi	6,467	0.609	0.315	0.344	0.065	49.8
1–2 mi	22,467	0.720	0.455	0.342	0.069	52.2
2–5 mi	62,237	0.827	0.584	0.284	0.094	54.6
5–10 mi	72,884	0.880	0.628	0.205	0.123	55.3
10+ mi	227,057	0.838	0.603	0.192	0.128	55.5

Note: Turnout figures are shares of all registered voters in each band, not conditional shares. Voters within 1 mile of SVH are noticeably younger, more Democratic, and had substantially lower 2022 turnout relative to their 2020 levels.

B.3 Table A12: Distance Effect on 2022 General Turnout Across Specifications

Table B15: Distance Coefficient on 2022 General Turnout Across Logistic Regression Specifications

	Model	Estimate	Std. Error	<i>p</i> -value
1	Raw (distance only)	+0.00806	0.000415	< 0.001
2	+ 2020 turnout	+0.00678	0.000465	< 0.001
3	+ Demographics	−0.00175	0.000483	< 0.001
4	+ Voter frequency	−0.000369	0.000525	0.482
5	Binary: within 2 mi (Model 3 controls)	−0.240	0.0148	< 0.001

Note: Outcome is an indicator for voting in the 2022 general election. All models are logistic regressions. “Demographics” (Model 3) adds party, age, age², sex, and race. “Voter frequency” (Model 4) adds indicators for prior election participation. The sign reversal from Models 1–2 to Model 3 reflects compositional differences: voters near SVH are younger and more Democratic, groups with lower baseline midterm turnout. $N = 394,150$.

B.4 Table A13: Full Logistic Regression — 2022 General Turnout with Demographics

Table B16: Full Logistic Regression: 2022 General Turnout (Model 3), $N = 394,150$

	Estimate	Std. Error	<i>z</i>	<i>p</i> -value
(Intercept)	−4.907	0.042	−115.68	< 0.001
Distance (miles)	−0.00175	0.000483	−3.62	< 0.001
Voted 2020	+2.608	0.013	196.03	< 0.001
Party: Other/Unenrolled	−0.075	0.009	−8.07	< 0.001
Party: Republican	+0.056	0.014	+4.09	< 0.001
Age	+0.064	0.001	48.54	< 0.001
Age ²	−0.000296	0.000012	−24.54	< 0.001
Female	−0.139	0.008	−18.29	< 0.001
Race: Black	+0.096	0.035	+2.77	0.006
Race: Hispanic	−0.295	0.026	−11.55	< 0.001
Race: Native American	+0.618	0.240	+2.58	0.010
Race: Unknown	+0.137	0.052	+2.61	0.009
Race: White	+0.748	0.021	35.51	< 0.001
Null deviance	533,782 on 394,149 df			
Residual deviance	423,908 on 394,137 df			
AIC	423,934			

Note: Reference categories: Democrat (party), Male (sex), Asian (race). Logistic regression via `glm(..., family = binomial)`.

B.5 Table A14: Subgroup Analysis — Retention vs. Mobilization

Table B17: Distance Effect by Subgroup: Retention (2020 Voters) vs. Mobilization (2020 Non-Voters)

Subgroup	N	Distance estimate	Std. Error	p -value
Retention: voted 2020, outcome = voted 2022	327,941	−0.00188	0.000510	< 0.001
Mobilization: did not vote 2020, outcome = voted 2022	66,209	+0.000597	0.00154	0.698

Note: Both models include party, age, age², sex, and race as controls. The distance effect is concentrated entirely among 2020 voters (retention channel); proximity had no statistically significant effect on mobilizing new voters.

B.6 Table A15: Robustness Checks — Distance Specification

Table B18: Robustness Checks: Alternative Distance Specifications for 2022 General Turnout

Model	Term	Estimate	Std. Error	p -value
Log distance	log(distance)	+0.0261	0.00445	< 0.001
Quadratic	Distance (linear)	−0.00120	0.000568	0.035
Quadratic	Distance ²	-7.62×10^{-6}	6.67×10^{-6}	0.253
Binary: within 1 mi	$\mathbf{1}[\text{dist} \leq 1]$	−0.430	0.0271	< 0.001
Binary: within 2 mi	$\mathbf{1}[\text{dist} \leq 2]$	−0.240	0.0148	< 0.001
Binary: within 5 mi	$\mathbf{1}[\text{dist} \leq 5]$	−0.0408	0.00921	< 0.001

Note: All models include 2020 turnout, party, age, age², sex, and race. The negative distance effect is robust across continuous, log, quadratic, and binary distance specifications. The effect decays monotonically: voters within 1 mile show the largest suppression (−0.430 log-odds), declining to −0.041 within 5 miles.

C Additional Voter-Level Analyses

C.1 Registration Trends Across Distance Thresholds

Figure C1 compares weekly registration shares across the matched 2017–2018 and 2021–2022 midterm cycles using four cumulative distance thresholds. The panels complement the main-text estimates by displaying the temporal pattern underlying each cutoff. Because the thresholds are cumulative and the panels use independent vertical scales, the figure is intended to compare timing within thresholds rather than levels across thresholds.

Registration trends across distance thresholds

Matched 2017–2018 and 2021–2022 midterm cycles

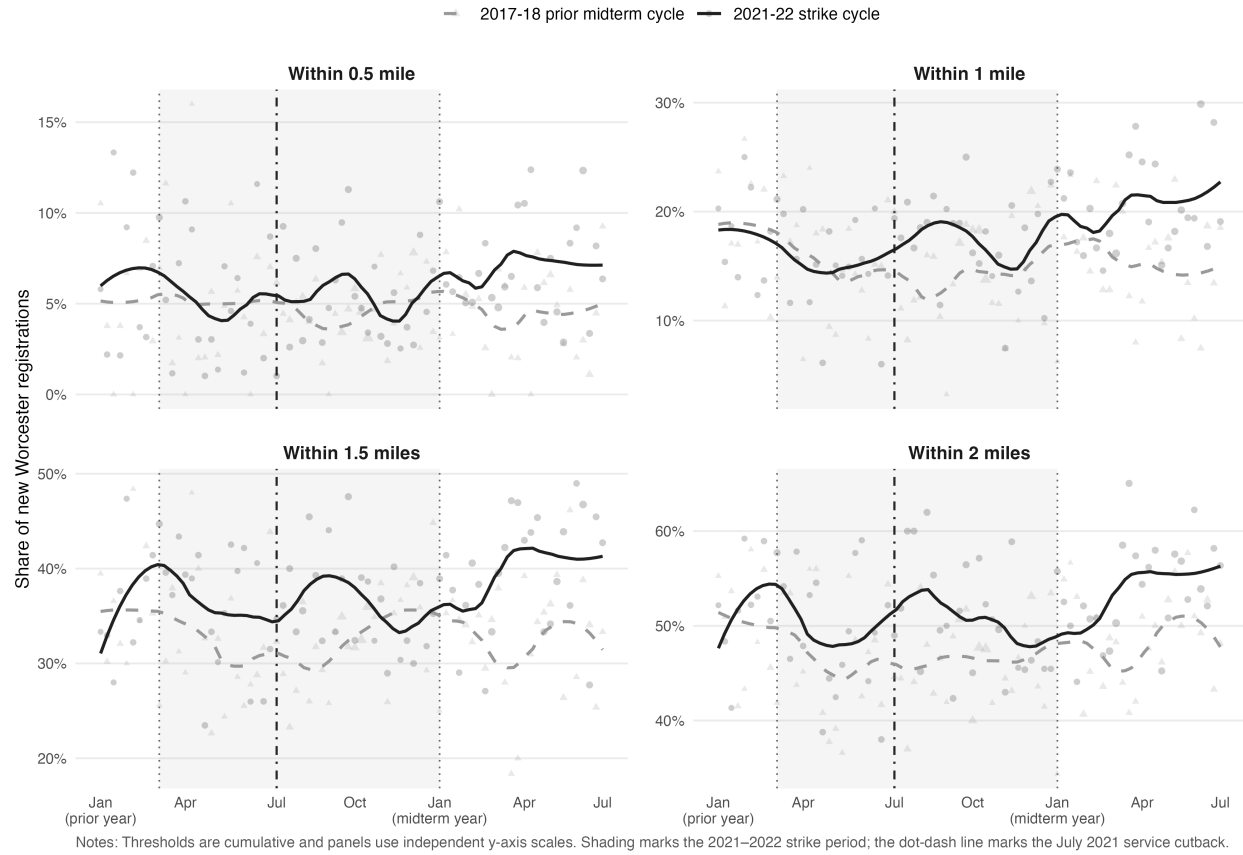


Figure C1: Matched-Cycle Registration Trends Across Distance Thresholds. Each panel shows the weekly share of all new Worcester registrants living within the specified distance of Saint Vincent Hospital. The solid line represents the 2021–2022 strike cycle and the dashed line the 2017–2018 prior midterm cycle. Shading marks the strike period; the dot-dash vertical line marks the July 2021 service cutback. Thresholds are cumulative, and panels use independent vertical scales.

C.2 Table A16: Worcester City Municipal Election Turnout

Table C19: Municipal Election Turnout, Worcester City Registered Voters ($N = 108,639$)

Election	Did not vote	Voted
2019 Municipal General	108,298	341
2021 Municipal General	108,358	281

Note: Counts are all Worcester city registered voters in the voter file, including those registered after 2020. Very low vote totals reflect that the voter file records participation for the entire city population, many of whom have no vote history entry for off-year municipal elections. These elections are not used in the main DiD analysis due to extremely low match rates to the voter file.

C.3 Table A17: Cross-Sectional Distance Effects, Worcester City

Table C20: Cross-Sectional Distance Effect on General Election Turnout, Worcester City Voters

	Election	Estimate	Std. Error	<i>p</i> -value
Placebo (pre-strike)	2018 General	+0.117	0.00893	< 0.001
Main outcome	2022 General	+0.118	0.00857	< 0.001

Note: Both models are unadjusted logistic regressions of turnout on distance from SVH. Sample restricted to Worcester city voters registered by October 2020 with non-missing age and party ($N = 71,153$). The positive cross-sectional coefficient reflects that within city limits, more distant voters tend to live in more suburban, higher-turnout neighborhoods. Cross-sectional estimates should not be interpreted causally; the DiD in Table A18 removes this baseline gradient.

C.4 Table A18: Alternative Two-Mile Difference-in-Differences, 2022 vs. 2018

Table C21: Alternative Two-Mile DiD: 2022 vs. 2018 General Turnout, Worcester City ($N = 71,153$)

Specification	Term	Estimate	Std. Error	<i>p</i> -value
Continuous distance	Distance (miles)	+0.157	0.00766	< 0.001
	Post-strike	-0.208	0.0265	< 0.001
	Distance $\times$ Post-strike	+0.0336	0.0107	0.002
Binary: within 2 mi	Within 2 mi	-0.316	0.0167	< 0.001
	Post-strike	-0.108	0.0156	< 0.001
	Within 2 mi $\times$ Post-strike	-0.0587	0.0235	0.012

Note: This supplementary analysis uses the broader cohort registered by 2020 and a two-mile threshold, rather than the fixed historical cohort and one-mile threshold used in the primary turnout analysis. Data are voter-election observations stacked across the 2018 and 2022 general elections. Models include party, age, age², sex, and race as controls. The binary specification (-0.059 log-odds, $p = 0.012$) indicates voters within 2 miles of SVH experienced a greater turnout decline from 2018 to 2022 relative to more distant Worcester City voters. The continuous distance interaction ($+0.034$, $p = 0.002$) implies the same pattern: each additional mile from SVH is associated with a relatively larger post-strike gain (i.e., proximity predicts greater post-strike suppression).

C.5 Table A19: Alternative Presidential-Cycle Placebo, 2020 vs. 2016

Table C22: Placebo DiD: 2020 vs. 2016 General Turnout, Worcester City (Binary within-2mi)

Term	Estimate	Std. Error	<i>p</i> -value
Within 2 mi	-0.341	0.0171	< 0.001
Post (2020)	+0.769	0.0178	< 0.001
Within 2 mi $\times$ Post	-0.103	0.0253	< 0.001

Note: This supplementary placebo uses the broader cohort and two-mile threshold from Table C21. It compares the 2020 and 2016 presidential elections, both occurring before the 2021–2022 nurses’ strike. The interaction term (-0.103 , $p < 0.001$) is statistically significant and larger than the main DiD estimate, suggesting that voters near SVH were already exhibiting differential turnout trends relative to more distant voters prior to the strike. This pre-existing pattern likely reflects compositional differences between near-SVH neighborhoods (younger, more Democratic) and city-wide averages across high- and low-salience election cycles. The post coefficient ($+0.769$) captures the large nationwide surge in 2020 presidential turnout relative to 2016. The significant placebo warrants caution in attributing the 2022 DiD effect solely to the strike.

C.6 Table A20: Heterogeneity in DiD Interaction Term

Table C23: Heterogeneous Treatment Effects: DiD Interaction (Within 2 mi $\times$ Post-Strike)

Subgroup	<i>N</i> (persons)	Estimate	Std. Error	<i>p</i> -value
Full sample (binary, no controls)	71,153	−0.0587	0.0235	0.012
<i>By party</i>				
Democrats only	—	−0.0281	0.0378	0.457
<i>By sex</i>				
Women	—	−0.0694	0.0300	0.021
Men	—	−0.0253	0.0309	0.413
<i>By age</i>				
Under 35	—	−0.173	0.0553	0.002
<i>By prior voting frequency</i>				
2016 non-voters (irregular)	—	−0.169	0.0462	< 0.001
2016 voters (regular)	—	−0.0480	0.0301	0.111
<i>By distance threshold</i>				
Within 0.5 mi $\times$ Post	—	−0.143	0.0598	0.017

Note: Each row presents the DiD interaction coefficient from a separate logistic regression using the binary distance indicator, estimated on the indicated subgroup or using the listed threshold. The full-sample model includes party, age, age², sex, and race; subgroup models omit the stratifying variable. The strongest effects appear among women (−0.069, $p = 0.021$), voters under 35 (−0.173, $p = 0.002$), and irregular voters who did not vote in 2016 (−0.169, $p < 0.001$). The effect is null for men, Democrats, and regular 2016 voters.

C.7 Table A21: Triple-Difference, Female $\times$ Near SVH $\times$ Post-Strike

Table C24: Triple-Difference Estimates: Female $\times$ Near SVH $\times$ Post-Strike

Distance threshold	Three-way interaction term	Estimate	Std. Error	<i>p</i> -value
≤ 0.5 mi	Within 0.5mi $\times$ Female $\times$ Post	−0.183	0.120	0.126
≤ 1.0 mi	Within 1mi $\times$ Female $\times$ Post	−0.0869	0.0711	0.221
≤ 1.5 mi	Within 1.5mi $\times$ Female $\times$ Post	−0.0604	0.0516	0.242
≤ 2.0 mi	Within 2mi $\times$ Female $\times$ Post	−0.0552	0.0470	0.240
≤ 3.0 mi	Within 3mi $\times$ Female $\times$ Post	−0.0610	0.0543	0.262
Continuous	Distance $\times$ Female $\times$ Post	+0.0252	0.0215	0.240

Note: Each row presents the three-way interaction coefficient from a separate logistic regression including all main effects, two-way interactions, party, age, age², and race. While the binary estimates are consistently negative (suggesting women near SVH had larger post-strike turnout declines relative to women further away), no specification reaches conventional significance levels. The data are estimated in the DiD stack of 2018 and 2022 elections.

C.8 Table A22: Voter Registration During the Strike Period

Table C25: Distance Effect on Probability of Registering During the Strike Period

Term	Estimate	Std. Error	<i>p</i> -value
Distance from SVH (miles)	+0.0269	0.0155	0.083

Note: Logistic regression of an indicator for registering during the strike period (March 8, 2021 – January 3, 2022) on distance from SVH, age, sex, and race. Sample restricted to Worcester city voters who registered between January 2020 and December 2022 (*N* limited to this registration window). The positive, marginally significant coefficient ($p = 0.083$) implies that more distant voters were somewhat more likely to register during the strike period, offering no evidence that proximity to SVH mobilized new voter registration.

C.9 Table A23: DiD by Precinct Racial Composition

Table C26: DiD Interaction by Precinct Racial Composition (Within 2 mi $\times$ Post-Strike)

Precinct subgroup	Estimate	Std. Error	<i>p</i> -value
Majority white ($\geq 50\%$ white)	−0.0443	0.0217	0.041
Majority nonwhite ($< 50\%$ white)	+0.00156	0.255	0.995
High white ($\geq 70\%$ white)	−0.369	0.412	0.369

Note: Precinct racial composition is computed from the voter file as the share of registered voters with race coded as White, aggregated to the precinct level. Each row presents the DiD interaction from a logistic regression estimated within the indicated precinct subgroup, without demographic controls. The negative effect is concentrated in majority-white precincts and is statistically significant (−0.044, $p = 0.041$). The null result in majority-nonwhite precincts ($p = 0.995$) is difficult to interpret given the large standard error. The high-white subgroup estimate is imprecise, reflecting sparse observations of within-2mi voters in precincts that are $\geq 70\%$ white.

C.10 Summary of Additional Voter-Level Findings

Tables A16–A23 present supplementary individual-level analyses using the Worcester city voter file. The alternative specification in Table A18 uses a broader cohort and a two-mile threshold and likewise shows a greater near-hospital turnout decline from 2018 to 2022 (−0.059 log-odds, $p = 0.012$). Heterogeneity analysis (Table A20) describes how that two-mile estimate varies across voter subgroups. The triple-difference (Table A21) does not find a statistically significant additional interaction for female voters.

The presidential-cycle placebo using the same broader cohort and two-mile threshold is also negative (Table A19, −0.103, $p < 0.001$), providing additional evidence that differential turnout trends near SVH predate the strike. These supplementary estimates reinforce the primary conclusion that the post-strike decline cannot be attributed to the strike.